

Probability: A Mathematical Introduction

Sample Academic Paper — Similarity Testing

Abstract

Probability is the mathematical study of uncertainty. It measures how likely an event is to occur and is used in games, weather forecasting, science, finance, and statistics.

Introduction

An experiment has a set of possible outcomes called the sample space. An event is a collection of one or more outcomes. The probability of an event is between zero and one.

Applications

For equally likely outcomes, probability can be calculated by dividing the number of favorable outcomes by the total number of possible outcomes. For example, a fair die has six outcomes, and the probability of getting an even number is $3/6$, or $1/2$.

Conclusion

Probability is useful because real-world decisions often involve uncertainty. It provides a numerical method for comparing possible outcomes and estimating risk.